From Heartbeat to Cardiochoreography: A Mechanics Foundation Model for Individualized 4D Cardiac Motion Generation Conditioned on Electrophysiology

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Abstract

The electrocardiogram (ECG) is a time series that records the electrical activity driving cardiac motion through electromechanical coupling (EMC), where the depolarization captured by ECG triggers each heartbeat. The temporal delay and low spatial resolution of such multichannel dynamic data often cause temporal and spatial misalignments along the physical motion, limiting the application of ECG data to event/segment-based downstream clinic practice. Although many efforts have been made to recover spatiotemporal trajectories conditioned on ECG waveforms, the intrinsic principle of EMC that results in various temporal delays remains underexplored in current generative models. To address this limitation, we propose Time2Track, a delay-aware generative model that generates individualized 4D heart motion from same-day ECG recordings by encoding electrophysiology as delay-aware token embeddings. Specifically, a learnable wavelet decomposition is employed to characterize the electromechanical delay, which is then formulated as a physics-based conditioning signal for motion diffusion. Time2Track generates 4D motion with individual-specific kinematics, avoiding the collapse to a population average, i.e., a common failure scenario when diffusing motions whose inter-subject differences are subtle. On $n = 40,577$ subjects from the UK Biobank, our Time2Track accurately synthesizes 4D cardiac trajectories from ECG and preserves inter-subject diversity, a regime in which generic motion diffusion baselines collapse to a population average.

1 Introduction

Electrophysiology signals, including electrocardiogram (ECG) and surface electromyography (sEMG) are the measurement of the coordinated electrical activity of excitable tissues. Electrophysiology data precedes the mechanical actions of human tissues, ruled by a phenomenon known as electromechanical coupling (EMC). Therefore, the heart's depolarization produces the QRS complex, three combined waves (Q, R, and S) representing the main spike in ECG, milliseconds before ventricular contraction, governed by calcium-mediated excitation and contraction coupling [6, 31]. In parallel, motor-unit action potentials in the forearm precede finger joint rotation by a measurable electromechanical delay [7, 24]. Although the challenge of reliable correspondence between electrical dynamics and geometric motion presents a compelling opportunity to faithfully map electrophysiology time series (ETS) to 4D (3D + time) meshes of the heart, the success of ETS-to-4D motion task would allow us to non-invasively recover organ- and limb-level kinesthetics from inexpensive, wearable sensors,

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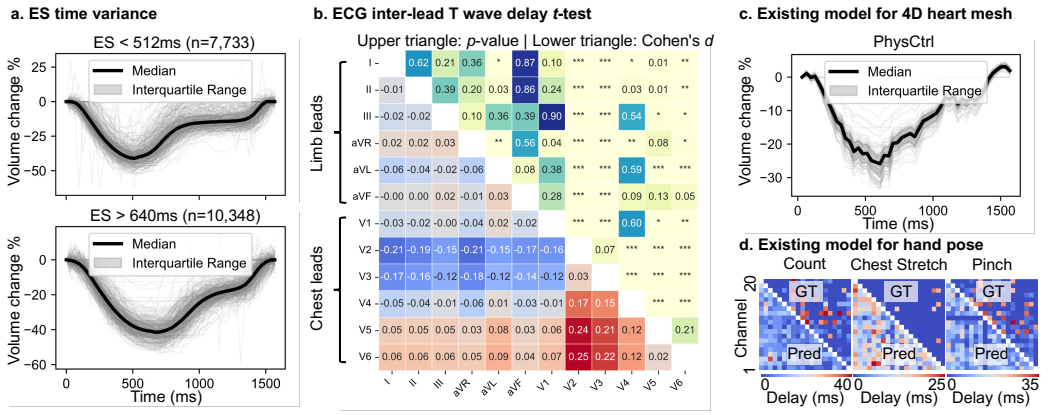


Figure 1: **(a)** The cardiac end-systolic (ES) stage occurred at different time across subjects, where the largest absolute volume change is used to determine the ES stage. **(b)** The ECG inter-lead T wave delay of subjects with ES occurred earlier than 512ms is compared with subjects entering the ES stage later than 640ms using t-test (*: 1e-3$\leq p <$1e-2, **: 1e-4$\leq p <$1e-3, ***: $p <$1e-4), where the delay is mainly observed between chest ECG leads, i.e., V2 and V3. **(c)** PhysCtrl failed on replicating the ES time variance inter-subject. **(d)** Hand pose delay is diverse by gesture, where the delay of ground truth (GT) hand joint angle cannot be replicated by NeuroPose predictions (Pred).

enabling continuous cardiac function monitoring, hand pose tracking, and large-scale phenotyping of movement and pathology.

Despite decades of progress, applications of electrophysiology remain limited for clinical practice. Most pipelines framed the problem into event detection [13], segment classification [3, 32, 2], or scalar regression [34], collapsing a rich spatiotemporal target into low-dimensional labels. The underlying reason is that electrophysiology data are fundamentally misaligned with the motion they encode since the electrical signal leads the mechanical response by a variable delay [6, 7]. Furthermore, the low spatial resolution of multichannel electrodes blurs the geometry of the source [5, 12]. These temporal and spatial misalignments hinder precise end-to-end 4D trajectory prediction.

An effective solution is to leverage recent spatiotemporal generative models for 3D trajectories, such as MeshHeart [29], PhysCtrl [42], and broader human motion diffusion frameworks [39, 52, 35]. However, these models are predominantly conditioned on static attributes and demographics [29, 47], anatomical priors [42], textual descriptions [39, 52], or univariate signals [40], rather than the dynamic, multivariate electrophysiology time series that actually derive motion. One natural attempt is to encode the raw electrophysiology sequence with a temporal backbone and frame it as a condition. However, we argue that this approach is not sufficient due to the ignorance of the multivariate structure of electrophysiology, in which each channel pair exhibits its own temporal delay and causal relationship to the resulting motion, and these relationships vary substantially across subjects and behaviors.

The most supporting physiological evidence is that the timing delay across ECG leads reflects the sequence of regional ventricular activation, and the resulting electrical and mechanical dys-synchrony is closely coupled together in deformation imaging [15]. Under this premise, subject-to-subject variability in inter-channel delays should manifest as corresponding variability in motion timing, i.e., an EMC fundamental that generative models should learn. Figure 1 quantifies this gap on two modalities. For cardiac EMC, we stratify UK Biobank [28] subjects, with 4D meshes derived via established cardiac segmentation and shape-modeling pipelines [4, 50], by the time at which the left ventricle reaches end-systole (ES), which is defined as the moment of largest absolute volume change (Fig. 1a). Such inter-subject variability in end-systolic timing is consistent with long-documented dispersion of systolic time intervals across healthy and diseased populations [48, 18]. Subjects reaching ES before 512ms and those reaching ES after 640ms exhibit significantly different inter-lead T-wave delays, with the effect mainly shown in the chest leads V2 and V3 (Fig. 1b), mirroring the known dispersion of ventricular repolarization captured by regional T-wave morphology [26, 27]. A state-of-the-art (SOTA) physics-conditioned generator, PhysCtrl [42], collapses this inter-subject ES timing variance, producing nearly identical volume-change trajectories regardless of the underlying ECG (Fig. 1c). On the other hand, a parallel pattern emerges for hand kinesthetics. The delay

between sEMG and joint-angle trajectories diverges across gestures, and predictions from NeuroPose [19] fail to reproduce the ground-truth delay distribution (Fig. 1d). Taken together, these observations support our argument that accurate 4D generation from electrophysiology requires a conditioning mechanism that is explicitly delay-aware across various spatial-temporal scales from multivariate ETS, rather than treating it as a generic temporal or multivariate embedding.

To that end, we introduce **Time2Track**, a delay-aware generative model of electromechanical coupling that generates 4D trajectories of individualized heart kinematics from electrophysiological dynamics. Time2Track differentiates from existing works in (1) a delay conditioning mechanism for EMC, developed on the heart and transferred to two sEMG decoders of hand pose in Sec. 4, and (2) a learnable delay modeling of multi-scale wavelets per channel pair. Unlike prior approaches that process electrophysiology through generic temporal backbones or global feature aggregation, Time2Track explicitly disentangles the representation of multivariate signals into distinct morphology and inter-channel delay branches. Existing sequence encoders tend to implicitly average out the fine-grained temporal phase shifts between channels, which explains why models like PhysCtrl and NeuroPose collapse the natural variance of electromechanical coupling into generic, averaged trajectories (as demonstrated in Fig. 1). Time2Track directly solves this spatial and temporal misalignment by explicitly extracting precise temporal offsets across multi-scale wavelets using Fast Fourier Transform Cross-Correlation (FFT-CC). By leveraging these raw, inter-channel time shifts into a high-dimensional, continuous space via a novel, window-aware Fourier feature lag embeddings, we preserve the critical timing differentials that dictate biological motion. Consequently, rather than relying on a black-box correlation, our motion diffusion backbone is conditioned on explicitly quantified physical delays alongside wavelet morphology. This allows the cross-attention mechanism to selectively attend to the exact activation wavefronts that drive mechanical responses, effectively recovering the subject-specific cardiac timing variance.

We summarize our contributions as follows: **(i) Learnable delay tokenization:** We propose a novel mechanism to explicitly extract and tokenize inter-channel delay across multi-scale wavelets from learnable decomposition. By using fast Fourier transform cross-correlation and proposing Fourier feature lag embeddings, these continuous temporal delays are embedded into a high-dimensional latent space. Time2Track preserves the fine-grained phase relationships that dictate individualized motion. **(ii) Delay-aware EMC generator:** We introduce Time2Track, the first delay-aware ETS framework that generates electrophysiology dynamics conditioning 4D motion. By explicitly embedding temporal delay and waveform morphology, it fundamentally aligns electrical signals with mechanical responses. **(iii) SOTA ETS-to-motion generation:** Evaluated on the largest benchmark of ECG-to-motion, over 40,000 combined instances across cardiac [28], Time2Track effectively bridges the gap between 1D biosignals and 4D kinematics, successfully recovering accurate, individual-specific motions from a rest-state prior and multivariate time series.

2 Related Work and Limitations

Deep Learning for Electromechanical Coupling. Modeling the EMC spans both computational physiology and data-driven machine learning. Traditionally, multiscale cardiac models couple ionic channel dynamics with active-stress generation and passive tissue mechanics to simulate the full cardiac cycle from a given action potential [31, 6]. While such physics-based approaches achieve high-fidelity virtual hearts, they depend on subject-specific parameterization via invasive or multimodal imaging and are computationally prohibitive at the population scale. In contrast, data-driven approaches have avoided these requirements at the cost of mechanistic interpretability. For instance, deep neural networks trained on ECG predict scalar indices such as ejection fraction, age, and arrhythmia labels [13, 32, 2], but reduce the 4D trajectory to classification or regression. Crucially, none of the aforementioned methods model the inter-channel delay structure of electrophysiology, nor do they serve as an explicit conditioning variable for predicting the continuous 4D motion of the target anatomy.

Representation Learning for Electrophysiological Signals. CoTAR [51] observes that electrophysiological signals typically originate from centralized biological sources, and replaces the decentralized "all-to-all" self-attention of standard Transformers with a centralized aggregation mechanism built around a learnable core token, as attention along the neural pathways between brain regions does for structural-functional coupling [45, 44]. However, this design does not explicitly capture the temporal relationships between events occurring across different channels. On the other hand, PhysioWave

[8] learns a softmax-weighted combination of wavelet templates across multiple decomposition scales within a BERT-style pretraining framework, yielding morphological tokens that are robust to noise and generalize across ECG and sEMG modalities, much as pretrained foundation models of brain fMRI generalize across cognitive tasks and cohorts [43, 46]. Yet, like CoTAR, PhysioWave encodes multichannel structure through joint attention over all channels, leaving the inter-channel phase relationships entangled, precisely the relationships that carry directional information about the underlying activation wavefront and drive inter-subject variability in kinematics. A broader line of work, Differentiable Digital Signal Processing (DDSP) [10], integrates classical signal-processing operators into end-to-end trainable networks, including parameterized band-pass filters [30] and learnable wavelet networks [21, 11, 33, 23]. These methods, however, target generic feature extraction rather than the modeling of electromechanical coupling. To the best of our knowledge, no prior work extracts inter-channel delays as a structured intermediate representation tied to event localization, nor integrates such delays with adaptive wavelet decomposition.

Conditional Diffusion Models for ECG and 4D Motion. Existing ECG diffusion work targets signal synthesis rather than motion, e.g., SSSD-ECG [1], DiffuSETS [16], ECGTwin [17], and SE-Diff [36]. Human motion diffusion models such as MDM [39], MotionDiffuse [52], PhysCtrl [42], and music-driven approaches like EDGE [40] synthesize articulated body motion conditioned on text descriptions, physical laws or univariate audio signals, but are not designed to accept multivariate electrophysiology as a conditioning input. Cardiac-specific generators, including MeshHeart [29], produce 4D heart mesh trajectories in exclusive conditions on static demographic features and anatomical shape priors rather than the dynamic ECG signal that mechanistically drives the motion. The critical and unaddressed gap is therefore a generative model that (i) accepts multivariate electrophysiology time series as its primary conditioning signal and (ii) outputs a continuous, individualized 4D motion, precisely the contribution of Time2Track.

3 Method

Time2Track takes a resting ECG and a single rest-state heart mesh and produces the individualized 4D trajectory. To prevent the collapse to the population average, we use a learnable multi-scale wavelet decomposition to form morphology and delay tokens for spatio-temporal diffusion conditioning.

Problem Definitions: This work aims to address the problem of trajectory generation in conditions of electrophysiological signals. Given a multivariate electrophysiological signal $\mathbf{e} \in \mathbb{R}^{C\times T}$ comprising C channels, and a subject-specific rest-state representation $\mathbf{p}_0 \in \mathbb{R}^{N\times S}$ (where N denotes the number of control points, with each point containing S features, e.g., for a 3D mesh, $S = 3$), we propose the Time2Track model. This model is designed to recover the trajectory of each control point as it evolves over time, specifically in the following form:

$$\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_F] \in \mathbb{R}^{F\times N\times S}, \mathbf{x}_f \in \mathbb{R}^{N\times S}. \tag{1}$$

Time2Track leverages a Denoising Diffusion Probabilistic Model (DDPM) [14] to model the conditional distribution $p(\mathbf{X}|\mathbf{e}, \mathbf{p}_0)$. Per DDPM, the model consists of forward process, which involves progressively adding Gaussian noise to $\mathbf{X}$ across diffusion time steps $k \in \{1, \ldots, K\}$ to generate $\mathbf{X}^{(k)} = \sqrt{\bar{\alpha}_k}\mathbf{X} + \sqrt{1-\bar{\alpha}_k}\boldsymbol{\epsilon}$, and a denoising network f_θ to predict the denoised trajectory.

$$\hat{\mathbf{X}} = f_\theta\big(\mathbf{X}^{(k)}, k, \mathbf{p}_0, \mathbf{c}(\mathbf{e})\big), \tag{2}$$

where $\mathbf{c}(\mathbf{e}) \in \mathbb{R}^{M\times D}$ denotes conditional tokens generated by our proposed electrophysiology encoder, comprising M tokens, each with a dimensionality of D. The model's training objective combines a motion reconstruction loss with the standard $\boldsymbol{\epsilon}$-prediction loss:

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{k,\boldsymbol{\epsilon},(\mathbf{X},\mathbf{e},\mathbf{p}_0)}\left[\left\|\mathbf{X} - f_\theta\big(\mathbf{X}^{(k)}, k, \mathbf{p}_0, \mathbf{c}(\mathbf{e})\big)\right\|_2^2\right]. \tag{3}$$

The core challenge lies in encoding two complementary types of features that are often overlooked in existing downstream tasks: (i) **Morphology**, specifically the shape characteristics of multiple transient waveforms across different scales, and (ii) **inter-channel delays**, referring to the temporal offsets between signals from different channels, which carry directional information regarding the underlying activation wavefront and the inter-subject difference of kinematic characteristics as illustrated in Fig. 1. This section focuses on our methods of electrophysiology time series encoding.

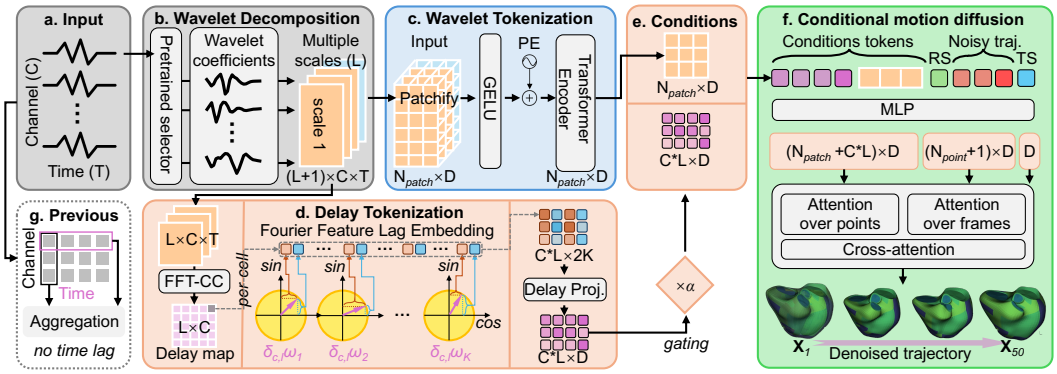


Figure 2: Time2Track framework. **(a)** The multivariate time series input. **(b)** The learnable wavelet decomposition produces the wavelet coefficients, $\mathbf{W} \in \mathbb{R}^{(L+1)\times C\times T_{\text{pad}}}$. The orange matrices denote detail scales blue is an approximation scale. **(c)** The wavelet branch first patchifies $\mathbf{W}$ to a sequence of windowed time series, and then it encodes tokens via Transformer encoder along with a positional embedding (PE). **(d)** The delay branch takes the detail scales $\mathbf{W}_{:L}$ and extracts temporal shift per (channel, scale) as a delay map $\Delta \in \mathbb{R}^{L\times C}$, followed by a conversion from delay value to a $2K$-long vector for tokenization. **(e)** Conditions of the motion diffusion are concatenated by wavelet and delay tokens, where delay tokens are gated by α. **(f)** Conditional motion diffusion generates a denoised heart mesh trajectory using a spatial-temporal Transformer given wavelet and delay tokens. **(g)** Although previous approaches encode aggregated features across channels and time, Time2Track produces delay tokens for the motion with only subtle and individual-specific differences.

3.1 Delay Aware Electrophysiology Encoder

The workflow of Time2Track is outlined in Fig. 2. It consists of two branches for the two aforementioned challenges, starting from (a) the multi-channel input and (b) a learnable multi-scale wavelet decomposition module. The morphology branch is mainly derived from PhysioWave [8], given the pretrained wavelet decomposition for ECG and EMG to produce robust morphology tokens in (c). The delay branch runs over decomposed wavelets and learns on the delay tokenization in (d). In the final, two branches are merged in (e) and applied in (f) for the conditional motion diffusion.

(a) Input. The raw signal is a z-normalized, C-channel signal segment $\mathbf{e} \in \mathbb{R}^{C\times T}$. For the morphology branch, $\mathbf{e}$ is zero-padded to a length of $T_{\text{pad}} = \lceil T/P \rceil P$, where P denotes the length of a temporal block. Whereas for the delay branch, we utilize the unpadded raw $\mathbf{e}$ to avoid spurious correlations introduced by zero-padding.

(b) Wavelet decomposition. $\mathbf{e}$ is then mapped by a pretrained wavelet decomposition model, PhysioWave, to a multi-scale spectrogram. The selector of PhysioWave mixes a bank of wavelet template $\{\psi_m\}_{m=1}^{M_\psi}$ via a softmax-weighted convex combination per scale, resulting in a L-level decomposition of detail coefficients $\{\mathbf{d}^{(\ell)}\}_{\ell=1}^{L}$ and an approximation $\mathbf{a}^{(L)}$. We stack them along the channel axis and denote it as $\mathbf{W}$ for the following wavelet tokenization:

$$\mathbf{W} = \text{Concat}\big(\mathbf{d}^{(1)}, \ldots, \mathbf{d}^{(L)}, \mathbf{a}^{(L)}\big) \in \mathbb{R}^{(L+1)\times C\times T_{\text{pad}}}. \tag{4}$$

(c) Morphology branch (wavelet tokenization). $\mathbf{W}$ is first patchified into $N_{\text{patch}} = (L+1)C\lceil T_{\text{pad}}/P \rceil$ time blocks (1D patches) over (channel, scale) using a convolutional layer Patch, followed by a GELU non-linearity and 2D positional embedding $\mathbf{PE}_{\text{2d}}$. Then, a stack of pre-norm Transformer blocks $\mathcal{T}_{\text{morph}}$ produces N_{patch} tokens

$$\mathbf{H}_{\text{morph}} = \mathcal{T}_{\text{morph}}\big(\text{GELU}(\text{Patch}(\mathbf{W})) + \mathbf{PE}_{\text{2d}}\big) \in \mathbb{R}^{N_{\text{patch}}\times D'}, \tag{5}$$

which we project to the latent width via a linear map: $\mathbf{T}_{\text{morph}} = MLP_{\text{morph}}(\mathbf{H}_{\text{morph}}) \in \mathbb{R}^{N_{\text{patch}}\times D}$.

(d) Delay branch (delay tokenization). The delay branch extracts the temporal shift between every channel and a reference channel r, at every detail scale of wavelet from $\mathbf{W}$, where only detail bands are used and denoted by $W_c^{(\ell)} \in \mathbb{R}^T$ with $\ell \in \{1, \ldots, L\}$ and channel $c \in \{1, \ldots, C\}$. The fast

Fourier transform cross correlation (FFT-CC) is utilized as shown in Fig. 3 to measure the delay. It first gets cross-correlation

$$R_c^{(\ell)}[\tau] = \frac{\mathcal{F}^{-1}\big(\widetilde{W}_c^{(\ell)} \odot \overline{\widetilde{W}_r^{(\ell)}}\big)[\tau]}{\|W_c^{(\ell)}\|_2 \|W_r^{(\ell)}\|_2 + \varepsilon}, \widetilde{W} = \mathcal{F}(W), \tag{6}$$

where $\mathcal{F}$ denotes FFT, and $\overline{\widetilde{W}_r^{(\ell)}}$ denotes element-wise complex conjugate of the reference spectrum, so that the element-wise multiplication of $\widetilde{W}_c^{(\ell)}$ and $\overline{\widetilde{W}_r^{(\ell)}}$ is the cross-spectrum and the inverse fast Fourier transform (IFFT) is the circular cross-correlation by the Wiener-Khinchin theorem [25].

We restrict the soft-argmax to a physically plausible window $\tau \in [-\tau_{\max}, \tau_{\max}]$ depending on the data modality, resulting in a temperature-controlled soft-argmax as follows:

$$\hat{\tau}_c^{(\ell)} = \sum_{\tau=-\tau_{\max}}^{\tau_{\max}} \tau \frac{\exp\big(\beta_\ell R_c^{(\ell)}[\tau]\big)}{\sum_{\tau'} \exp\big(\beta_\ell R_c^{(\ell)}[\tau']\big)}, \beta_\ell = \mathrm{softplus}(\beta_\ell^{\mathrm{raw}}) + \epsilon. \tag{7}$$

where $\beta_\ell^{\mathrm{raw}} \in \mathbb{R}$ is a learnable scalar parameter. Therefore, inter-channel delay can be defined by $\delta_c^{(\ell)} = \hat{\tau}_c^{(\ell)} - \hat{\tau}_r^{(\ell)}$, which obviously exhibits shift invariance in $\mathbf{e}$. Afterward, the resulting delay map $\mathbf{\Delta} \in \mathbb{R}^{C \times L}$ is tokenized via a Fourier feature lag embedding approach.

Fourier feature lag embedding. Continuous lags are notoriously hard for MLPs to extract features. We therefore lift each $\delta_c^{(\ell)}$ to a sinusoidal embedding analogous to neural radiance field (NeRF) positional encodings [22], but window-aware: We use K log-spaced angular frequencies tied to the search window, $\omega_k = \frac{2^k \pi}{\tau_{\max}}$, $k = 0, \ldots, K-1$, yielding a $2K$-dimensional feature

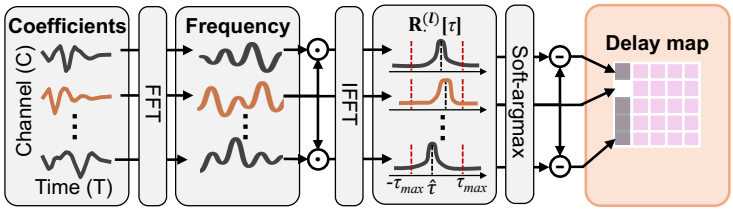


Figure 3: The workflow of fast Fourier transform cross correlation (FFT-CC) to extract the delay map from decomposed wavelet coefficients, where the orange channel denotes the reference channel, $\odot$ denotes element-wise multiplication, and $R_c^{(\ell)}$ refers to Eq. 6. The reference channel is applied $\odot$ and $\ominus$ to itself.

$$\boldsymbol{\phi}\big(\delta_c^{(\ell)}\big) = \big[\sin(\omega_0\delta), \cos(\omega_0\delta), \ldots, \sin(\omega_{K-1}\delta), \cos(\omega_{K-1}\delta)\big]. \tag{8}$$

The sine/cosine clock plot in Fig. 2d vividly illustrates this multi-resolution lifting process. As the dimensionality of inter-channel time lag is lifted, the delay projector can robustly map each $\boldsymbol{\phi}(\delta_c^{(\ell)}) \in \mathbb{R}^{2K}$ to $\mathbb{R}^D$. Note that channel and scale embeddings, which are omitted in Fig. 2d for clarity, $\mathbf{u}_c, \mathbf{v}_\ell \in \mathbb{R}^D$ are also accumulated to every token of the delay map during the delay projection

$$\mathbf{T}_{\mathrm{delay}} = \alpha \cdot \mathcal{T}_{\mathrm{delay}}\big([g_\phi\big(\boldsymbol{\phi}(\delta_c^{(\ell)})\big) + \mathbf{u}_c + \mathbf{v}_\ell]_{c,\ell}\big) \in \mathbb{R}^{C \cdot L \times D}, \tag{9}$$

where the learnable scalar parameter $\alpha \in \mathbb{R}$ is zero initialized at the start of training, so that the encoder is stable at the beginning to just fine-tune the wavelet decomposition.

(e) Condition tokens. The morphology and delay tokens are concatenated as the condition tokens, $\mathbf{c}(\mathbf{e}) = \mathrm{Concat}\big(\mathbf{T}_{\mathrm{delay}}, \mathbf{T}_{\mathrm{morph}}\big) \in \mathbb{R}^{(C \cdot L + N_{\mathrm{patch}}) \times D}$, in which the type and ordering are retrievable through the channel, scale, and positional embeddings. We deliberately avoid pooling either branch so that cross-attention in Fig. 2f can attend selectively to morphological wavelets or to a specific delay token at channel c, scale ℓ.

3.2 Electrophysiology Conditional Motion Diffusion

The electrophysiology conditional Motion Diffusion Model (MDM) shown in Fig. 2f is a Spatio-Temporal Transformer, the same as general MDMs [38, 42], for kinematic trajectory. First, the rest-state (RS) representation $\mathbf{p}_0$ (green token in Fig. 2f) is prepended to $\mathbf{X}$ as $\widetilde{\mathbf{X}} \in \mathbb{R}^{(F+1) \times N \times S}$ so that the initial pose is attended. Then, it is mapped to the latents of 4D motion denoted by

$\mathbf{Z} \in \mathbb{R}^{(F+1)\times N\times D}$ via an MLP projector. The backbone interleavedly proceeds spatial self-attention over the N points within a frame and temporal self-attention over the $F+1$ frames for the same point, with the diffusion timestep (TS, blue token in Fig. 2f) injected via Adaptive Layer Normalization (AdaLN), updating $\mathbf{Z}$ every layer. Conditioning on the input signal is implemented by cross-attention

$$\mathbf{Z}'_{f,n} = \mathbf{Z}_{f,n} + \mathrm{CrossAttn}\big(\mathbf{Z}_{f,n}, \mathbf{c}(\mathbf{e})\big). \tag{10}$$

After L_{dit} layers, an MLP head (omitted in Fig. 2) reads off the predicted trajectory offset $\hat{\mathbf{X}}$ with respect with the RS slot

$$\hat{\mathbf{X}} = \hat{\mathbf{X}}_{1:F} + \mathbf{p}_0, \tag{11}$$

which improves convergence by enforcing the trivial trajectory as a cheap prior. At inference, we run DDIM sampling [37] for K_{infer} steps and apply classifier-free guidance.

4 Experimental Results

Datasets. The experiments of this work involve two applications, ECG conditional heart motion generation using UKB [28]. UKB contains large subject and kinematic variability that has demonstrated EMC diversity in Fig. 1, resulting in a challenging task for general motion diffusion models and existing electrophysiology encoders without inter-channel delay consideration. UKB has 53,224 subjects that span the age range from adults to the elderly. The 4D heart mesh is extracted from ECG-gated cardiac MRI for one cycle, and quality controlled, released by [41]. and it is represented by the 388×3 control points. After pairing with resting-state 12-lead ECG (500Hz) captured at the same visit as MRI and filtering samples with missing frames, there are $n = 40,577$ ECG-mesh pairs. Electrophysiology-motion pairs are time aligned, and the train:validation ratio is 80:20%. Since the protocol has no held-out test split, every checkpoint is selected by the training objective itself, the one-step DDPM denoising loss, so that no fidelity or diversity metric enters model selection, and all encoders share one training configuration without hyperparameter search.

Baselines and implementations. To our knowledge, no prior diffusion model conditions 4D motion on multi-variate electrophysiology signal as reviewed in Section 2. Therefore, we hold the diffusion backbone based on a general-purpose state-of-the-art (SOTA) MDM, PhysCtrl, and vary the electrophysiology encoder, so that experimental results isolate the contribution of the conditioning representation. The diffusion backbone is a Spatial-Temporal Transformer with 8 layers, 256 latent channels, 8 conditioning tokens, 50 DDIM inference steps, etc., as the default settings of PhysCtrl. We also evaluated MDM to provide a sanity check for a generic motion-diffusion approach on this task. Encoders chosen for ECG-mesh data: **(i) CNN**: a vanilla 1D-CNN encoder over the 12-lead waveform that emits 8 conditioning tokens, serving as a minimal baseline. **(ii) TeCh** [51]: a temporal-channel transformer that used 2-layer and 2-layer centralized self-attention over channel and time, respectively, producing an SOTA fused ECG token. **(iii) PhysioWave** [8]: a BERT-style wavelet foundation model, mixed 4 types of wavelets, embed dim 384, depth 8, 12 heads, and initialized from publicly released ECG pretrained weights. **(iv) ShiftInvariant** [9]: a shift invariant model capturing inter-lead timing implicitly without any external annotation. **(v) Delay-T**: the CNN encoder augmented with an explicit 12×12 inter-lead T-peak delay matrix that is encoded by a per-row MLP (hidden 128) into 12 additional conditioning tokens, providing direct supervision of a manually selected delay type. **(vi) Delay-QRS**: identical to Delay-T, but the delay of the QRS complex is manually selected. Table 6 in Appendix A.4 summarizes the architecture, the handling of time lag and the size of every encoder.

Metrics. The measurement of experimental results focuses on both global and local structures, trends, and absolute value of location, speed, and surface area, as well as the similarity of inter-subject diversity between the predicted and true 4D motion. We contrast predicted (Pred) motion against ground truth (GT) along complementary axes as follows: **(i) Global trend** is measured by the per-subject **RV coefficient** and Rooted Mean Squared Error (**RMSE**) between the raw Pred and GT trajectories, capturing whether the model recovers the dominant systolic-diastolic pattern across the cycle, which is ruled by EMC. **(ii) Local trend.** After cubic detrending of each curve to suppress the global trend, **PAR=**$(max(\hat{\text{Pred}}) - min(\hat{\text{Pred}}))/(max(\hat{\text{GT}}) - min(\hat{\text{GT}}))$ reports the ratio between peak-to-peak magnitudes of detrended predicted ($\hat{\text{Pred}}$) and GT ($\hat{\text{GT}}$) trajectories. PAR quantifies 'does it put local peaks at the right size'. **(iii) Inter-subject diversity** is the main measurement in this work, designed to quantify whether a model collapses to a population mean when the inter-subject

Table 1: Whole-heart 3D vertex trajectory metrics (Pred vs. GT). Inter-subject diversity columns are reported as $\widehat{\text{Pred}}/\widehat{\text{GT}}$ ratios, with $\pm$ the SD over the 10,000 paired subject bootstrap resamples described in Appendix A.5.

Method	RV Coefficient↑	RMSE↓	PAR↑	$\lvert\text{APCR} - 1\rvert \downarrow$	$\lvert\text{APDR} - 1\rvert \downarrow$
CNN	0.634±0.150	0.130±0.046	0.466±0.106	0.241±0.058	0.151±0.007
TeCh	0.636±0.153	**0.129±0.045**	0.401±0.098	0.356±0.065	0.170±0.012
PhysioWave	0.634±0.174	0.172±0.074	0.868±0.090	0.044±0.037	0.079±0.006
ShiftInvariant	0.558±0.168	0.144±0.052	0.729±0.249	0.452±0.078	0.305±0.009
Delay-T	0.619±0.177	0.158±0.057	0.772±0.082	0.042±0.046	0.091±0.008
Delay-QRS	0.589±0.179	0.166±0.061	0.894±0.145	0.322±0.049	0.025±0.006
Ours	**0.637±0.176**	0.151±0.069	**0.896±0.130**	**0.034±0.042**	**0.019±0.008**

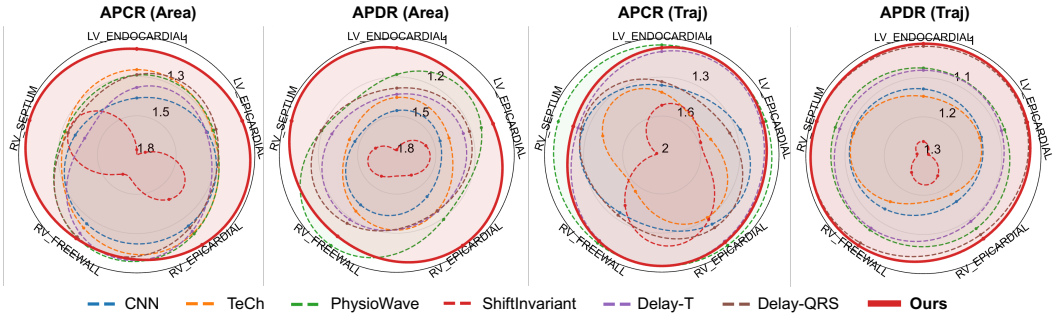


Figure 4: Radar charts of inter-subject diversity metrics, including Average Pairwise Coefficient Ratio (APCR) and Average Pairwise Distance Ratio (APDR) evaluated on mesh face area (Area) and vertex trajectory (Traj). The metrics are computed across five distinct anatomical regions of the heart preprocessed by [41], where RV/LV denotes Right/Left Ventricular. The outermost line of each radar represents the ideal ratio of 1.0, meaning the predicted diversity perfectly matches the ground truth.

differences in 4D motion are subtle. We report the average pairwise distance ratio (**APDR**) and average pairwise coefficient ratio (**APCR**) on the 3D vertex trajectory (**Traj**), the 1D mesh face area (**Area**), and the 1D speed of the vertex trajectory (**Speed**). For each metric, we compute the subject-to-subject distance or similarity coefficient (RV for 3D Traj, Pearson for 1D) on detrended predicted and GT curves, with detrending applied to remove the dominant global heartbeat pattern. APDR and APCR are then the ratios of the predicted to the GT values. Ratios far from one indicate collapse to a subject-invariant amplitude or slope. These diversity metrics are essential because UKB exhibits strong inter-subject variability (Fig. 1a), and a model that produces nearly identical motions for every subject can otherwise score deceptively well on global-trend metrics.

Quantitative Results: Table 1 presents the quantitative evaluation of the predicted whole-heart 3D vertex trajectories against the GT. While centralized encoders like CNN and TeCh achieve competitive RMSE scores (0.130 and 0.129, respectively), their significantly low PAR values (0.466 and 0.401) indicate an inability to match peak-to-peak local magnitudes. Furthermore, they exhibit high errors in inter-subject diversity metrics ($\lvert\text{APCR} - 1\rvert$ and $\lvert\text{APDR} - 1\rvert$), reflecting a severe collapse where the model predicts an averaged heart motion for all subjects without the subtle but individual-specific differences. In contrast, Time2Track achieves the best performance in capturing both the global structure and local trends, yielding the highest RV Coefficient (0.637) and PAR (0.896). Crucially, our method dominates in preserving inter-subject diversity, obtaining the lowest $\lvert\text{APCR} - 1\rvert$ (0.034) and $\lvert\text{APDR} - 1\rvert$ (0.019), which are nearly $10\times$ better than vanilla baselines. While explicitly supervising certain delays (Delay-T, Delay-QRS) or using foundation models (PhysioWave) improves diversity over the CNN baseline, they still fall short of our approach with the learnable delay.

These whole-heart observations are further supported by the fine-grained regional diversity analysis in the radar charts (Fig. 4). Across five distinct anatomical regions and across both face area (Area) and vertex trajectory (Traj) metrics, our method (solid red line) consistently stays closest to the ideal ratio of 1.0. Almost all baseline methods systematically underestimate this diversity, drawing inward toward the center of the radar charts, confirming that they tend to generate smoothed and

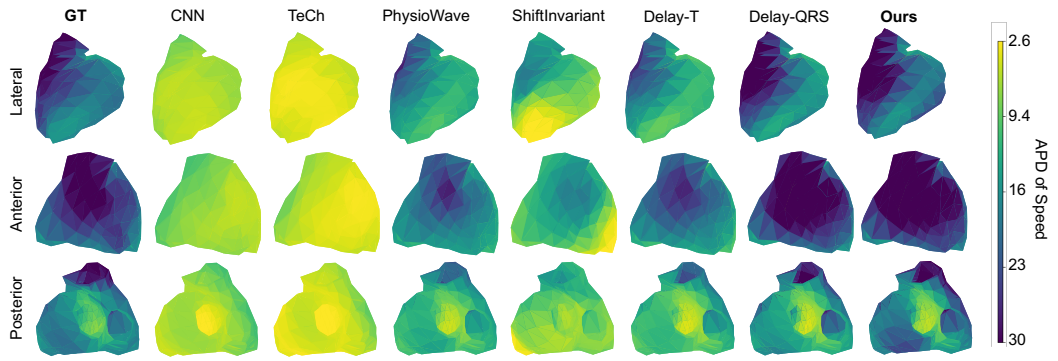


Figure 5: Qualitative visualization of inter-subject diversity, measured by the Average Pairwise Distance (APD) of vertex speed mapped onto the 3D heart meshes. The meshes are shown from three anatomical perspectives: Lateral, Anterior, and Posterior. The colormap indicates the mean of per-vertex APD across the validated subjects, where the prediction is expected to reproduce GT.

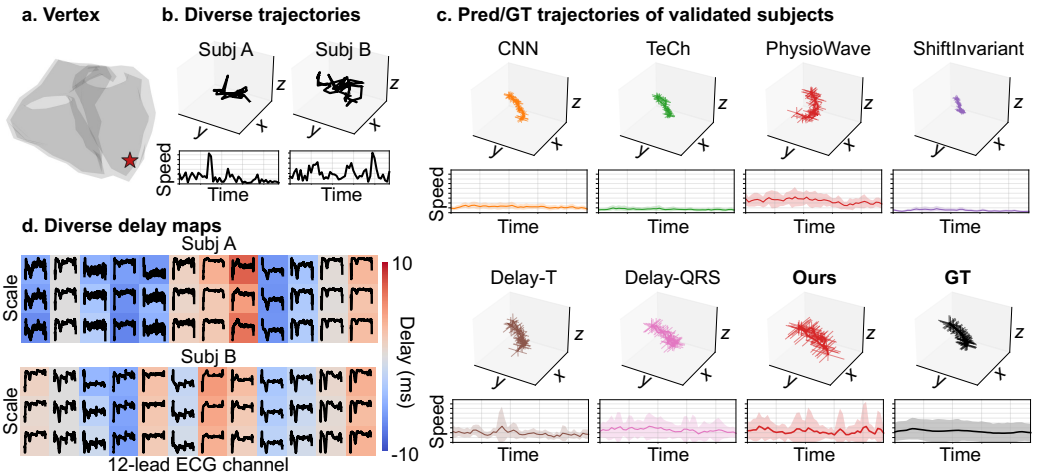


Figure 6: Qualitative comparison of trajectories and delays. **(a)** Randomly selected vertex on the right ventricle used here. **(b)** GT 3D trajectories and speed profiles for two subjects show high natural kinematic diversity. **(c)** Predicted 3D trajectories and speed distributions (mean curve and variance bar) across the validation set. Baselines suffer from severe mode collapse, predicting near-mean motions with near-zero variance. Conversely, Ours accurately recovers the spatial spread and speed variance, closely matching GT. **(d)** Extracted multi-scale inter-channel delay maps with wavelet coefficient curves. These subject-specific temporal shifts drive our model's ability to generate personalized motions instead of collapsing to the average.

averaged cardiac motions. The ablations in Appendix A.3 show that removing the delay branch raises $|\text{APDR} - 1|$ from 0.019 to 0.079 and that FFT-CC and the lag embedding outperform their simpler replacements.

Qualitative Results: Fig. 5 visualizes inter-subject diversity by mapping the Average Pairwise Distance (APD) of vertex speed onto the 3D heart meshes. In contrast, baseline encoders (CNN, TeCh, ShiftInvariant) produce low-variance, yellowish surfaces, failing entirely to capture patient-specific speed dynamics. While PhysioWave and explicit delay encoders show slight improvements, Time2Track is the best at reproducing the spatially heterogeneous APD distribution of the GT.

Furthermore, Fig. 6 highlights trajectory distributions for a randomly selected vertex on the right ventricle. As shown in Fig. 6b, true subject motions (e.g., Subj A vs. Subj B) exhibit highly diverse 3D paths and speed profiles. In Fig. 6c, baseline models collapse to a dense, near-mean trajectory bundle with nearly flat, zero-variance speed distributions. In contrast, Time2Track recovers wide variance bars and larger shade in speed plots, demonstrating a close mirror of GT. As illustrated in

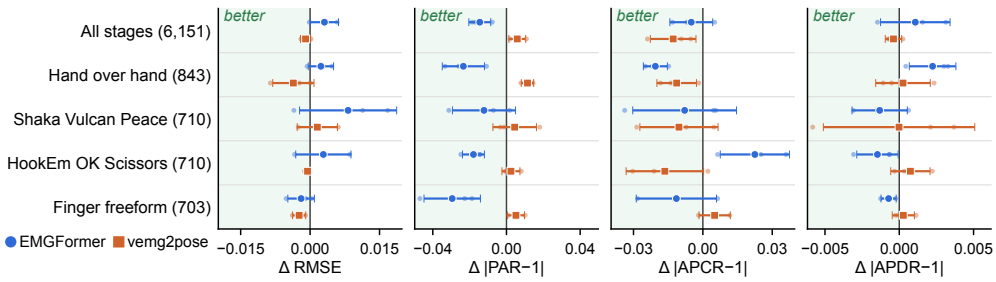


Figure 7: Change of each metric from adding delay tokens to the sEMG decoders EMGFormer [49] (circles) and vemg2pose [34] (squares) on the emg2pose test split (sessions in parentheses). Dots show three seeds, bars their mean ± SD, and shading the gains.

Fig. 6d, this superior generative capability is driven by our extracted multi-scale inter-channel delay maps.

Pathology and Robustness: We have compared the 981 validation subjects with a cardiac ICD-10 code with as many randomly sampled healthy subjects. Table 2 shows that Time2Track has the smallest RMSE change and the only decrease of $|\text{APCR} - 1|$, which rises by 0.034 for CNN and 0.064 for TeCh. The robustness tests in Appendix A.6 show little effect of electrode noise and τ_{max}, whereas dropping lead II raises $|\text{APCR} - 1|$.

Table 2: Diseased minus matched healthy subjects ($n = 981$, lower is better).

Δ	RMSE	$\lvert\text{PAR} - 1\rvert$	$\lvert\text{APCR} - 1\rvert$	$\lvert\text{APDR} - 1\rvert$
Ours	**+0.001**	+0.009	**−0.025**	+0.004
CNN	+0.005	**−0.007**	+0.034	−0.001
TeCh	+0.005	+0.008	+0.064	**−0.005**

ECG as the Only Input: Replacing the subject's rest mesh $\mathbf{p}_0$ with the population average $\hat{\mathbf{p}}_0$ at inference, without retraining, leaves a resting 12-lead ECG as the only input for 4D heart generation without cardiac imaging. Table 3 shows that the cost falls on fidelity (RV −0.046, RMSE +0.031) rather than on diversity.

Table 3: Own ($\mathbf{p}_0$) vs. population ($\hat{\mathbf{p}}_0$) rest mesh.

	$\mathbf{p}_0$	$\hat{\mathbf{p}}_0$	Δ
RV↑	0.637	0.591	−0.046
PAR → 1	0.896	0.867	−0.029
$\lvert\text{APCR} - 1\rvert$	0.034	0.052	+0.018
$\lvert\text{APDR} - 1\rvert$	0.019	0.036	+0.017
RMSE↓	0.151	0.182	+0.031

Transfer of Delay Tokens to sEMG: We inject the delay tokens into the sEMG hand pose decoders vemg2pose [34] and EMGFormer [49], trained with three seeds on the first gesture cycle of each emg2pose benchmark session (6,151 test sessions, 178 users). Fig. 7 shows that delay tokens lower $|\text{PAR} - 1|$ of EMGFormer in every subset (by 0.012 to 0.030) and $|\text{APCR} - 1|$ of both decoders in four of the five subsets, while RMSE and $|\text{APDR} - 1|$ change by less than 0.01 and $|\text{PAR} - 1|$ of vemg2pose rises by at most 0.011.

5 Conclusions and Limitations

In this work, we introduced Time2Track, a delay-aware generative model of electromechanical coupling that generates individualized 4D cardiac kinematics directly from multivariate electrophysiology time series. By explicitly modeling learnable inter-channel temporal delays through FFT cross-correlation and Fourier feature lag embeddings, our method effectively overcomes the severe collapse observed in generic motion diffusion models with various existing ECG encoders, successfully recovering natural inter-subject kinematic diversity. With the rest mesh from an existing cardiac MR scan, a resting ECG replaces the ECG-gated cine acquisition for motion assessment. Despite these advances, our approach has three limitations. First, it depends on the searching window size τ_{max} for cross-correlation, which may restrict its applicability in extreme cases with highly abnormal electromechanical delays. Second, the UKB imaging cohort is about 90% healthy, with diagnoses from registry codes rather than adjudicated imaging, so severe or acute pathology remains untested. Third, the 388 control points are coarser than the ventricular wall thickness, so regional wall-motion abnormalities lie beyond the reach of our model and its metrics. Future work will focus on higher-resolution representations, validation in clinical cohorts, and real-time motion tracking in severe cardiovascular and neuromuscular disorders.

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A Appendix

A.1 Dataset and Code Access

The heart mesh and ECG data can be accessed in the UK Biobank after applying for permission at this URL https://www.ukbiobank.ac.uk/. Code can be accessed at this URL https://github.com/Chrisa142857/time2track

A.2 Implementation Details and Hyperparameters

Time2Track is implemented in PyTorch with HuggingFace `accelerate`. Unless stated otherwise, all baselines share the diffusion backbone (Spatio-Temporal Transformer, $L_{\text{dit}} = 8$ layers, latent width $D = 256$, $N_{\text{patch}}{+}C{\cdot}L = 48$ conditioning tokens) and only the electrophysiology encoder is varied, so observed differences are attributable to the conditioning representation. Training and inference settings are summarized in Table 4.

α-gating schedule and wavelet fine-tuning. The delay-branch gate α is a single learnable scalar initialized to 0, so at step 0 the model reduces exactly to the PhysioWave morphology baseline. α is then updated freely by AdamW (no manual schedule) and grows as the delay tokens become useful. The wavelet front-end is initialized from the publicly released PhysioWave ECG checkpoint and not frozen: every parameter (template mixing weights, learnable kernels, and the soft-argmax temperatures $\{\beta_\ell^{\text{raw}}\}$) is updated end-to-end with the diffusion loss.

Dataset preprocessing. Resting 12-lead ECG (500Hz) is cropped to a single 600-sample window aligned to the cardiac MRI cycle and z-normalized per channel. Cardiac meshes have $N{=}388$ control points and are sampled to $F{=}50$ frames per cycle. Subjects with missing frames are filtered, yielding $n{=}40{,}577$ ECG/mesh pairs split 80/20 train/val with seed 42.

A.3 Ablation Studies

We isolate the contribution of the delay branch and of the FFT-CC delay extractor by holding the diffusion backbone and training schedule of Sec. A.2 fixed and varying only the electrophysiology encoder. Table 5 reports whole-heart 3D vertex metrics on the UKB validation set. The $|\text{APCR} - 1|$ and $|\text{APDR}-1|$ columns directly measure inter-subject diversity collapse. All numbers are reproduced from the corresponding rows of Table 1 for direct comparison.

Table 4: Training and model hyperparameters for Time2Track on UKB ECG→4D heart mesh.

Optimization	
Optimizer / (β_1, β_2) / ϵ	AdamW / $(0.9, 0.999)$ / 10^{-8}
Weight decay	1×10^{-2}
Learning rate / schedule / warmup	1×10^{-5} / cosine / 200 steps
Per-GPU batch size / grad-accum / effective batch	20 / 4 / 320 (4×GPUs)
Epochs / max grad norm / mixed precision	200 / 1.0 / bf16
Dropout / cond. drop rate (CFG train)	0.1 / 0.1
Gradient checkpointing / xFormers attn.	enabled / enabled
Diffusion	
Backbone / layers / latent dim D	ST-Transformer / 8 / 256
Inference sampler / steps K_{infer} / CFG scale	DDIM / 50 / 2.0
Trajectory frames F / control points N	50 / 388
Velocity loss weight λ_{vel}	1.0
Morphology branch	
Levels L / wavelet kernel size	3 / 24
Wavelet template $\{\psi_m\}$ [8]	db4, db6, sym4, coif2
Patch length P / morph. embed dim D'	64 / 384
Depth / heads	8 / 12
Init. weights / fine-tuning	[8] / full fine-tune
Delay branch	
Reference lead r	lead II
Search window τ_{max} (samples / ms @500Hz)	$\pm$100ms
Soft-argmax temperature init. β_ℓ^{raw}	softplus(2.0)
Fourier lag bands K / freqs. ω_k	8 / log-spaced, $2^k\pi/\tau_{\text{max}}$
Delay transformer heads	4
Gate α initialization	0
Hardware / runtime	
GPUs / wall-clock time	4×NVIDIA H100 NVL / ≈25h to convergence

Effect of the delay branch. Removing the delay branch costs roughly $4\times$ on $|\text{APDR} - 1|$ (0.019→0.079), confirming that delay tokens are the primary lever against collapsinng to the average.

Effect of FFT-CC. Replacing FFT-CC with a single T-peak or QRS-peak delay (Delay-T / Delay-QRS) shows only minor improvement of the inter-subject diversity and loses local fidelity (PAR drops from 0.896 to 0.772 for Delay-T).

Skip the FFT-CC and use a directly regressed scalar lag per (channel, scale) pair (→ *raw scalar lag*) leaves worse point-wise error and inflates inter-subject diversity (|APCR-1| 0.081 vs. 0.034) and shrinks |APDR-1| by 0.195. The differentiable cross-correlation provides a sharper, less noisy lag signal than direct regression.

Effect of the lag embedding. Both removing the Fourier-feature embedding (feeding the scalar lag through an MLP) and removing positional embeddings $\mathbf{u}, \mathbf{v}$ on the delay tokens have a worse RV and APDR than the full Time2Track, confirming that the NeRF-style $\sin / \cos$ bands and channel/scale positions both carry signal that the backbone cannot recover from a scalar.

Effect of patch length for the morphology branch. Varying the patch length (4 / 8 / 16) *collapses* amplitude and inter-subject diverisity (PAR ≤ 0.726, APDR ≤ 0.472, APCR ≥ 2.275). The full model's bank of {db4, db6, sym4, coif2} with 64 patches strikes the right multi-resolution coverage; both under- and over-parameterized banks lose the electromechanical phase information that the delay branch is meant to inject.

Table 5: Component ablations on UKB whole-heart vertex trajectories. "Ours" is the full Time2Track implemented with hyperparameters listed in Table 4.

Variant	RV ↓	RMSE ↓	PAR → 1	APCR → 1	APDR → 1
Delay branch					
w/o delay branch	0.634	0.172	0.868	0.956	0.826
FFT-CC					
→ Delay-T	0.619	0.158	0.772	1.042	0.743
→ Delay-QRS	0.589	0.166	0.894	1.042	0.907
→ raw scalar lag	0.615	0.178	0.887	1.081	0.786
Fourier-feature Lag embedding					
→ scalar + MLP	0.597	0.179	**0.896**	1.084	0.923
w/o $\mathbf{u}, \mathbf{v}$	0.595	0.160	0.858	1.042	0.790
Morphology branch					
Patch length (P=4)	0.588	**0.142**	0.275	2.398	0.139
Patch length (P=8)	0.512	0.144	0.726	2.275	0.472
Patch length (P=16)	0.545	0.145	0.276	2.391	0.144
Ours (full)	**0.637**	0.151	**0.896**	**0.966**	**0.981**

A.4 Encoder Properties

Table 6 lists the criterion for including every encoder, and its column of time lag covers no lag, shift invariance, classical T-peak and QRS delays, and the learned lag. Compared with the diversity in Table 1, the parameter count does not order the encoders, whereas the handling of time lag does: $|\text{APDR} - 1|$ is 0.151 and 0.170 without lag, 0.079 with wavelet scales, and 0.019 with the learned lag.

Table 6: Properties of the compared electrophysiology encoders.

Encoder	Architecture	Time lag	Why included	Params
CNN	1D conv on raw time	none	vanilla	0.21 M
TeCh	attention over time patches and leads	none	SOTA MedTS Transformer	8.16 M
PhysioWave	attention over (scale, time), wavelets	none	SOTA pretrained ECG	14.35 M
ShiftInvariant	shift invariant net, no attention	invariant	SOTA shift invariance	0.15 M
Delay-T	TeCh plus external delay matrix	classical T peak	classical delay	9.21 M
Delay-QRS	TeCh plus external delay matrix	classical QRS	classical delay	9.21 M
Ours	attention, learnable wavelets, delay	learned, gated	proposed	15.41 M

A.5 Error Bars of the Diversity Metrics

APCR and APDR are pairwise statistics over the diversity sample of 128 subjects, so their uncertainty is estimated by a paired subject-level bootstrap with 10,000 resamples, in which the numerator and the denominator of each ratio are recomputed on the same resample. The $\pm$ of $|\text{APCR} - 1|$ and $|\text{APDR} - 1|$ in Table 1 is the SD of $|\text{APCR} - 1|$ and $|\text{APDR} - 1|$ across the resamples, which ranges from 0.037 to 0.078 and from 0.006 to 0.012, respectively.

A.6 Robustness to Electrode Noise, Missing Leads and τ_{max}

Table 7 reports two tests on the diseased subgroup, per-lead Gaussian noise added at inference with σ scaled from the power of each lead to reach the target SNR, and lead channels set to zero at inference. Dropping the reference lead II leaves fidelity intact but raises $|\text{APCR} - 1|$, and the model still generalizes when three chest leads are dropped. Table 8 shows that a $4\times$ change of τ_{max} at inference keeps RV and RMSE within their bootstrap intervals and changes PAR, $|\text{APCR} - 1|$ and $|\text{APDR} - 1|$ by at most 0.011.

Table 7: Electrode noise and lead dropouts at inference on the diseased subgroup.

Input	RV↑	RMSE↓	PAR $\to 1$	$\lvert\text{APCR} - 1\rvert \downarrow$	$\lvert\text{APDR} - 1\rvert \downarrow$
clean	0.576	0.179	0.892	0.027	0.014
noise, 20 dB SNR	0.576	0.179	0.891	0.027	0.014
noise, 10 dB SNR	0.572	0.180	0.882	0.020	0.012
noise, 5 dB SNR	0.562	0.183	0.871	0.002	0.010
drop V2 (single chest)	0.579	0.178	0.888	0.024	0.017
drop II (reference)	0.575	0.183	0.958	0.108	0.011
drop V1, V2, V3 (three chest)	0.581	0.177	0.891	0.050	0.021

Table 8: Sensitivity to τ_{max} at inference, default in bold (95% bootstrap intervals for RV and RMSE).

τ_{max}	RV↑	RMSE↓	PAR $\to 1$	$\lvert\text{APCR} - 1\rvert \downarrow$	$\lvert\text{APDR} - 1\rvert \downarrow$
25 ($\pm$50 ms)	0.570 [0.557, 0.585]	0.181 [0.177, 0.185]	0.898	0.052	0.011
50 ($\pm$100 ms)	0.568 [0.555, 0.581]	0.181 [0.177, 0.186]	0.902	0.050	0.011
75 ($\pm$150 ms)	0.567 [0.552, 0.581]	0.181 [0.177, 0.186]	0.901	0.041	0.012
100 ($\pm$200 ms)	0.565 [0.552, 0.579]	0.181 [0.177, 0.186]	0.903	0.046	0.012

A.7 Reference Lead

Lead II is the standard reference lead in clinical electrocardiography owing to its high SNR [20]. Table 9 shows that retraining Time2Track with aVR or aVL as the reference lowers RV and raises RMSE, $\lvert\text{APCR} - 1\rvert$ and $\lvert\text{APDR} - 1\rvert$. In a preliminary ES timing classification without a reference lead, aggregating the full delay map by its mean or median reaches an AUC of 0.590 or 0.563, against 0.597 for the lead II delay. The median inter-lead T-peak delay in UKB is 8 ms (interquartile range 2 to 30 ms), so the 2 ms sampling interval at 500 Hz does not limit its quantification.

Table 9: Sensitivity to the reference lead.

Reference lead	RV↑	RMSE↓	PAR $\to 1$	$\lvert\text{APCR} - 1\rvert \downarrow$	$\lvert\text{APDR} - 1\rvert \downarrow$
II (Table 1)	0.637	0.151	0.896	0.034	0.019
aVR	0.590	0.181	0.948	0.041	0.022
aVL	0.592	0.181	0.934	0.074	0.024

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